

COMP3011 Coursework 1

API Documentation

Disaster Event Intelligence API

Version	Coursework submission version
Framework	FastAPI
Primary format	JSON
Base URLs	Local: http://127.0.0.1:8000 Deployment: https://jihaomarvin.pythonanywhere.com
Github Repo URLs	https://github.com/MarvinJihao/COMP3011_HaoJi

1. Overview

The Disaster Event Intelligence API is a SQL-backed disaster event intelligence service designed for coursework demonstration and technical evaluation. It supports local CRUD operations, external data ingestion, metadata tracking, analytical reporting, and system health monitoring.

The service is implemented with FastAPI and returns JSON for all operational endpoints. It is structured as a REST-style API with clear route grouping for events, ingestion, analytics, source metadata, and health checking.

- CRUD operations for locally stored disaster events
- Ingestion of wildfire data from NASA EONET
- Ingestion of earthquake data from the USGS Earthquake Catalog
- Metadata endpoints for registered sources and ingestion history
- Analytical endpoints for summary statistics, daily trends, hotspots, and risk assessment
- Health-check endpoints for service and database monitoring

The core persistence model is built around three main tables: disaster_events, source_metadata, and ingest_runs.

2. Authentication

Protected endpoints use HTTP Basic Authentication. This keeps the implementation straightforward for coursework demonstration while still showing route protection and access control.

Route group	Authentication
/events	Required
/ingest	Required
/analytics	Required
/sources	Required
/health	Not required
/health/db	Not required

Default coursework credentials are shown below and are intended only for demonstration and marking.

Username	Password
admin	admin123

Authorization header example

```
Authorization: Basic YWRtaW46YWRtaW4xMjM=
```

3. Common Status Codes

Status code	Meaning
200 OK	Successful request
201 Created	Resource created successfully
204 No Content	Resource deleted successfully
401 Unauthorized	Missing or invalid Basic Authentication credentials
404 Not Found	Requested resource does not exist
422 Unprocessable Entity	Validation or input error
502 Bad Gateway	External upstream API failure
503 Service Unavailable	Database health-check failure

4. Core Data Models

4.1 DisasterEvent

Field	Type	Description
id	integer	Internal primary key
title	string	Event title
type	string	Event type, for example wildfire or earthquake
latitude	number	Latitude in decimal degrees
longitude	number	Longitude in decimal degrees
severity	integer	Severity score in the range 1-5
source	string	Upstream or manual source key
external_id	string or null	Upstream event identifier when available
event_time	datetime	Event time in ISO 8601 format
created_at	datetime	Record creation time

4.2 SourceMetadata

Field	Type	Description
id	integer	Internal primary key
source_key	string	Short source identifier
title	string	Display title
provider	string	Provider name

source_url	string or null	Optional source URL
description	string or null	Optional source description
active	boolean	Whether the source is active
created_at	datetime	Record creation time

4.3 IngestRun

Field	Type	Description
id	integer	Internal primary key
provider	string	Upstream provider name
dataset	string	Dataset name
status	string	Job status, for example completed or failed
requested_count	integer	Number of external items returned
inserted_count	integer	Number of new records inserted
skipped_count	integer	Number of duplicates skipped
failed_count	integer	Number of items that failed processing
filters_json	string or null	Serialized sync parameters
message	string or null	Optional error or run message
started_at	datetime	Job start time
finished_at	datetime or null	Job completion time

5. Events Endpoints

5.1 Create Event

Property	Value
Endpoint	POST /events
Description	Creates a new disaster event record.
Authentication	Required

Request body

```
{
  "title": "Test Wildfire",
  "type": "wildfire",
  "latitude": 34.05,
  "longitude": -118.25,
  "severity": 3,
  "source": "manual",
```

```
{
  "external_id": "demo-001",
  "event_time": "2026-03-12T10:00:00"
}
```

Successful response - 201 Created

```
{
  "title": "Test Wildfire",
  "type": "wildfire",
  "latitude": 34.05,
  "longitude": -118.25,
  "severity": 3,
  "source": "manual",
  "event_time": "2026-03-12T10:00:00",
  "id": 1,
  "external_id": "demo-001",
  "created_at": "2026-03-12T10:00:03.191092"
}
```

5.2 List Events

Property	Value
Endpoint	GET /events
Description	Returns stored disaster events. Supports filtering and pagination.
Authentication	Required

Query parameters

Name	Type	Required	Description
skip	integer	No	Pagination offset, default 0
limit	integer	No	Max returned items, default 100, max 500
type	string	No	Filter by event type
source	string	No	Filter by source key
severity_min	integer	No	Minimum severity
severity_max	integer	No	Maximum severity
start_time	datetime	No	Lower bound for event time
end_time	datetime	No	Upper bound for event time
min_lat	number	No	Minimum latitude
max_lat	number	No	Maximum latitude
min_lon	number	No	Minimum longitude
max_lon	number	No	Maximum longitude

Example request

```
GET /events?type=wildfire&source=InciWeb&severity_min=2&limit=20
```

Successful response - 200 OK

```
[
  {
    "title": "Wildfire Example",
    "type": "wildfire",
    "latitude": 40.0,
    "longitude": -120.0,
    "severity": 3,
    "source": "InciWeb",
    "event_time": "2026-03-10T12:00:00",
    "id": 12,
    "external_id": "EONET_123",
    "created_at": "2026-03-10T12:01:00"
  }
]
```

5.3 Get Event By ID

Property	Value
Endpoint	GET /events/{event_id}
Description	Returns a single event by internal identifier.
Authentication	Required

Path parameters

Name	Type	Description
event_id	integer	Internal event identifier

Successful response - 200 OK

```
{
  "title": "Wildfire Example",
  "type": "wildfire",
  "latitude": 40.0,
  "longitude": -120.0,
  "severity": 3,
  "source": "InciWeb",
  "event_time": "2026-03-10T12:00:00",
  "id": 12,
  "external_id": "EONET_123",
  "created_at": "2026-03-10T12:01:00"
}
```

Error response - 404 Not Found

```
{
  "detail": "Event not found"
}
```

5.4 Replace Event

Property	Value
Endpoint	PUT /events/{event_id}
Description	Fully replaces an existing event.
Authentication	Required

The request body format is the same as POST /events. The endpoint returns the fully updated event object with status code 200 OK.

5.5 Patch Event

Property	Value
Endpoint	PATCH /events/{event_id}
Description	Partially updates an existing event.
Authentication	Required

Example request body

```
{
  "severity": 5,
  "source": "manual-review"
}
```

Successful response: 200 OK. The endpoint returns the updated event object.

5.6 Delete Event

Property	Value
Endpoint	DELETE /events/{event_id}
Description	Deletes an event by ID.
Authentication	Required

Successful response: 204 No Content. No response body is returned.

6. Ingestion Endpoints

6.1 Preview EONET Wildfires

Property	Value
Endpoint	GET /ingest/eonet/wildfires/preview
Description	Retrieves a preview of wildfire events from NASA EONET without writing to the database.

Authentication	Required
----------------	----------

Query parameters

Name	Type	Required	Description
days	integer	No	Number of recent days, default 7
limit	integer	No	Number of items to preview, default 20
status_filter	string	No	open, closed, or all

Successful response - 200 OK

```
{
  "count": 2,
  "events": [
    {
      "id": "EONET_1",
      "title": "Wildfire Example",
      "event_time": "2026-03-10T00:00:00",
      "latitude": 40.0,
      "longitude": -120.0,
      "source": "InciWeb"
    }
  ]
}
```

6.2 Sync EONET Wildfires

Property	Value
Endpoint	POST /ingest/eonet/wildfires/sync
Description	Fetches wildfire records from NASA EONET and stores new events in the local database.
Authentication	Required

Query parameters

Name	Type	Required	Description
days	integer	No	Number of recent days, default 30
limit	integer	No	Maximum external items, default 200
status_filter	string	No	open, closed, or all

Successful response - 200 OK

```
{
  "provider": "NASA EONET v3",
  "category": "wildfires",
}
```



```
"requested": 5,
"inserted": 4,
"skipped_existing": 1,
"failed": 0,
"filters": {
  "category": "wildfires",
  "days": 30,
  "limit": 200,
  "status": "open"
}
```

Error response - 502 Bad Gateway

```
{
  "detail": "Failed to fetch NASA EONET data: connection failed"
}
```

6.3 Sync USGS Earthquakes

Property	Value
Endpoint	POST /ingest/usgs/earthquakes/sync
Description	Fetches earthquake records from the USGS Earthquake Catalog and stores new events in the local database.
Authentication	Required

Query parameters

Name	Type	Required	Description
days	integer	No	Number of recent days, default 30
min_magnitude	number	No	Minimum magnitude, default 2.5
limit	integer	No	Maximum external items, default 500

Successful response - 200 OK

```
{
  "provider": "USGS FDSN Event API",
  "event_type": "earthquake",
  "requested": 10,
  "inserted": 8,
  "skipped_existing": 2,
  "failed": 0,
  "filters": {
    "format": "geojson",
    "starttime": "2026-02-11T12:00:00Z",
    "endtime": "2026-03-12T12:00:00Z",
    "minmagnitude": 2.5,
    "limit": 500,
    "orderby": "time"
  }
}
```

6.4 List Sources

Property	Value
Endpoint	GET /sources
Description	Returns registered source metadata stored locally.
Authentication	Required

Successful response - 200 OK

```
[
  {
    "id": 1,
    "source_key": "InciWeb",
    "title": "InciWeb",
    "provider": "NASA EONET v3",
    "source_url": null,
    "description": null,
    "active": true,
    "created_at": "2026-03-12T10:00:00"
  }
]
```

6.5 List Ingestion Runs

Property	Value
Endpoint	GET /ingest/runs
Description	Returns synchronization history.
Authentication	Required

Query parameters

Name	Type	Required	Description
provider	string	No	Filter by provider
dataset	string	No	Filter by dataset
limit	integer	No	Max returned runs, default 50

Successful response - 200 OK

```
[
  {
    "id": 1,
    "provider": "NASA EONET v3",
    "dataset": "wildfires",
    "status": "completed",
    "requested_count": 5,
    "inserted_count": 4,
    "skipped_count": 1,
  }
]
```

```

    "failed_count": 0,
    "filters_json": "{\"category\": \"wildfires\", \"days\": 30, \"limit\": 200, \"status\": \"open\"}\",
    "message": null,
    "started_at": "2026-03-12T10:00:00",
    "finished_at": "2026-03-12T10:00:04"
  }
]

```

7. Analytics Endpoints

7.1 Analytics Summary

Property	Value
Endpoint	GET /analytics/summary
Description	Returns aggregated event statistics.
Authentication	Required
Key query parameters	start_date date; end_date date

Successful response - 200 OK

```

{
  "total_events": 5,
  "time_range": {
    "start_date": null,
    "end_date": null,
    "oldest_event_time": "2026-03-08T19:26:00",
    "latest_event_time": "2026-03-10T22:40:00"
  },
  "by_type": [
    {
      "type": "wildfire",
      "count": 5
    }
  ],
  "by_source": [
    {
      "source": "IRWIN",
      "count": 5
    }
  ],
  "by_severity": [
    {
      "severity": 3,
      "count": 5
    }
  ]
}

```

7.2 Daily Timeseries

Property	Value
Endpoint	GET /analytics/timeseries/daily

Description	Returns daily counts of stored events.
Authentication	Required
Key query parameters	start_date date; end_date date

Successful response - 200 OK

```
{
  "series": [
    {
      "date": "2026-03-01",
      "count": 2
    },
    {
      "date": "2026-03-02",
      "count": 1
    }
  ]
}
```

7.3 Hotspots

Property	Value
Endpoint	GET /analytics/hotspots
Description	Returns coordinate clusters based on rounded latitude and longitude buckets.
Authentication	Required
Key query parameters	precision integer; top_n integer; start_date date; end_date date

Successful response - 200 OK

```
{
  "hotspots": [
    {
      "latitude": 35.1,
      "longitude": -120.1,
      "count": 2
    },
    {
      "latitude": 10.0,
      "longitude": 100.0,
      "count": 1
    }
  ]
}
```

7.4 Risk Assessment

Property	Value
Endpoint	GET /analytics/risk-assessment
Description	Ranks stored events using a composite risk score based on severity,

	event type, recency, and local event density.
Authentication	Required
Key query parameters	days integer; top_n integer; event_type string

Successful response - 200 OK

```
{
  "items": [
    {
      "event_id": 10,
      "title": "Recent Earthquake Cluster A",
      "type": "earthquake",
      "source": "us",
      "severity": 5,
      "event_time": "2026-03-16T11:00:00",
      "risk_score": 87.0,
      "risk_level": "critical",
      "factors": {
        "severity": 40.0,
        "type_weight": 30.0,
        "recency": 11.0,
        "local_density": 6.0
      }
    }
  ]
}
```

8. Health Endpoints

8.1 Health Check

Property	Value
Endpoint	GET /health
Description	Returns service availability status.
Authentication	Not required

Successful response - 200 OK

```
{
  "status": "ok",
  "service": "disaster-event-intelligence-api"
}
```

8.2 Database Health Check

Property	Value
Endpoint	GET /health/db
Description	Checks database connectivity.
Authentication	Not required

Successful response - 200 OK

```
{
  "status": "ok",
  "database": "connected"
}
```

Error response - 503 Service Unavailable

```
{
  "detail": "Database connection failed"
}
```

9. Validation and Error Format

Validation errors follow FastAPI's standard error structure.

Example validation error - 422 Unprocessable Entity

```
{
  "detail": [
    {
      "loc": ["body", "severity"],
      "msg": "Input should be less than or equal to 5",
      "type": "less_than_equal"
    }
  ]
}
```

10. Quick Verification Flow

- GET /health
- GET /health/db
- POST /ingest/eonet/wildfires/sync
- POST /ingest/usgs/earthquakes/sync
- GET /sources
- GET /ingest/runs
- GET /events
- POST /events
- GET /analytics/summary
- GET /analytics/timeseries/daily
- GET /analytics/hotspots
- GET /analytics/risk-assessment

11. Notes

- All protected endpoints require HTTP Basic Authentication.
- All timestamps are handled using ISO 8601 datetime strings.
- External ingestion depends on upstream service availability.
- The database used in this coursework implementation is SQLite.